

Centre Number	Candidate Number	Candidate Name
---------------	------------------	----------------

NAMIBIA SENIOR SECONDARY CERTIFICATE

BIOLOGY ORDINARY LEVEL

6116/3

PAPER 3 Alternative to Practical

1 hour 15 minutes

Marks 40

2024

Additional Materials: Ruler

Non-programmable calculator

INSTRUCTIONS AND INFORMATION TO CANDIDATES

- Candidates answer on the Question Paper in the spaces provided.
- Write your Centre Number, Candidate Number and Name in the spaces at the top of this page.
- Write in dark blue or black pen.
- You may use a soft pencil for any diagrams, graphs or rough working.
- Do not use correction fluid.
- You may use a non-programmable calculator.
- Do not write in the margin *For Examiner's Use*.
- Answer **all** questions.
- The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's use	
1	
2	
3	
4	

Total	
-------	--

Marker	
Checker	

This document consists of **12** printed pages and **4** blank pages.



Republic of Namibia

MINISTRY OF EDUCATION, ARTS AND CULTURE

- 1 (a) Describe how to investigate the effect of distilled water and salt solution in red onion epidermal cells.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

[5]

- (b) Fig.1.1 shows the image of the cells before salt solution was added.

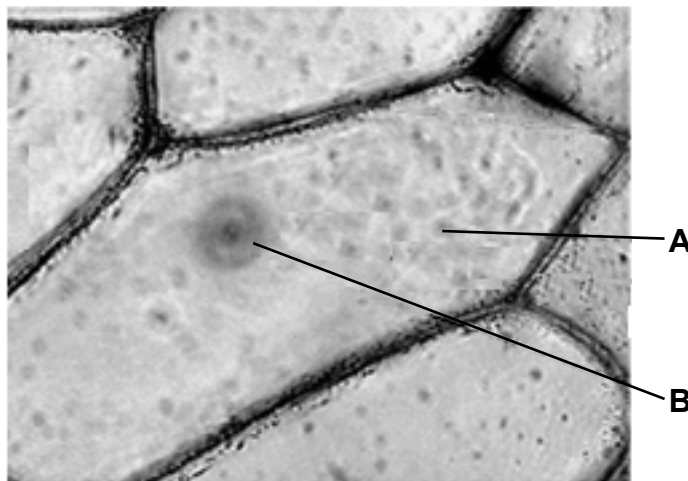


Fig. 1.1

- (i) Identify the parts labelled **A** and **B**.

A

B

[2]

Fig. 1.2 shows an image of the onion tissue after it was left in the salt solution.

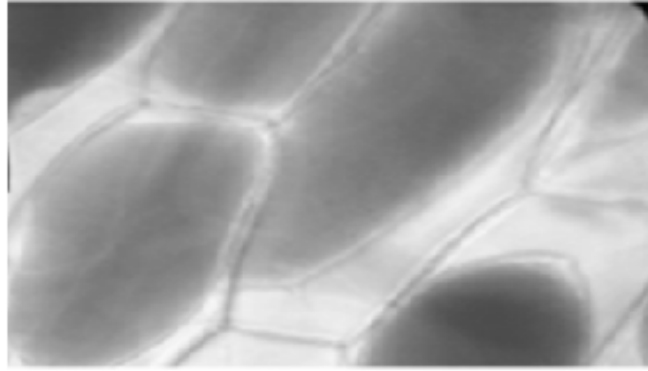


Fig. 1.2

- (ii) Describe and explain the changes observed in Fig. 1.2 after the onion tissue was left in the salt solution.

.....

.....

.....

.....

.....

.....

[3]

[10]

- 2** An experiment was carried out to investigate the effect of light on gas exchange of an aquatic plant leaf using hydrogencarbonate indicator solution.

Hydrogencarbonate indicator solution can be used to study gas exchange in different light conditions and will change from orange/red to yellow with increasing carbon dioxide or purple with decreasing carbon dioxide.

Hydrogencarbonate indicator solution was added to each of the 4 test-tubes. The apparatus was set up as follows:

- A** no leaf in the test-tube (control test-tube)
- B** a leaf was placed in the test-tube and left in the light
- C** a leaf was placed in the test-tube which was wrapped in black paper to block out the light
- D** a leaf was placed in the test-tube which was wrapped in gauze to allow partial light

A bung was placed in the mouth of each test-tube and all 4 test-tubes were left in the light for one hour.

Fig. 2.1 shows the 4 test-tubes.

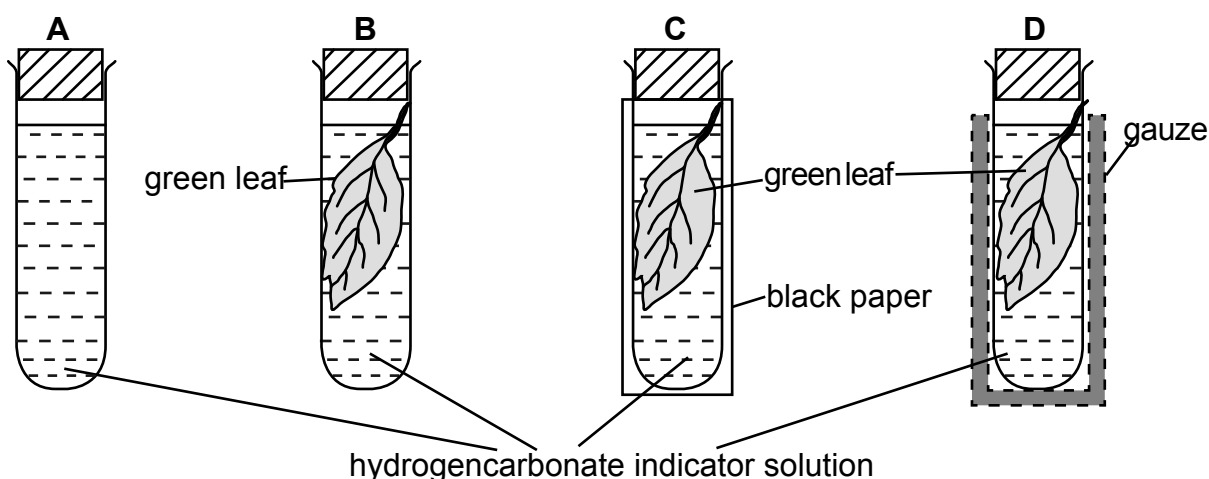


Fig. 2.1

Table 2.1 shows the results after one hour.

Table 2.1

test-tube	colour of hydrogencarbonate indicator at start	colour of hydrogencarbonate indicator after one hour
A	orange	orange
B	orange	purple
C	orange	yellow
D	orange	orange

(a) Suggest why the hydrogencarbonate indicator solution did not change colour in

(i) tube-tube **A**,

.....
..... [1]

(ii) test-tube **D**.

.....
.....
.....
..... [2]

(b) Explain the change in colour of the hydrogencarbonate indicator solution in

(i) test-tube **B**,

.....
..... [1]

(ii) test-tube **C**.

.....
.....
.....
..... [2]

- (c) Limewater is an indicator that can be used to show the presence of carbon dioxide.

Suggest why limewater would not be a suitable indicator for use in this investigation.

.....
.....

[1]

- (d) Give **two** variables that should be kept constant during this investigation.

1

2

[2]

- (e) State the purpose of the control test-tube in this investigation.

.....
.....

[1]

[10]

- 3 Fig. 3.1 shows a measuring cylinder containing water. Three learners recorded the volume reading from the measuring cylinder in Fig. 3.1. Their readings were:

- Learner 1: 98 cm^3
- Learner 2: 99 cm^3
- Learner 3: 100 cm^3

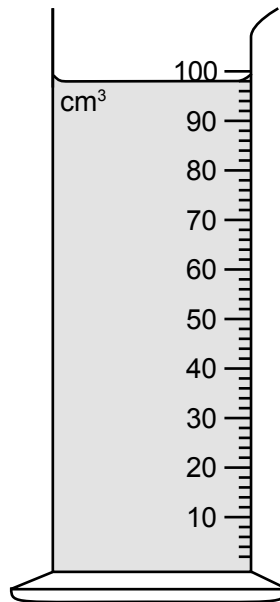


Fig. 3.1

- (a) (i) State which learner made an accurate reading.

..... [1]

- (ii) State the mistake made by **one** of the other two learners.

.....
 [1]

- (b) Grade 11 learners investigated the rate of cooling of a warm body using beakers containing hot water under different conditions.

Fig. 3.2 shows the apparatus.

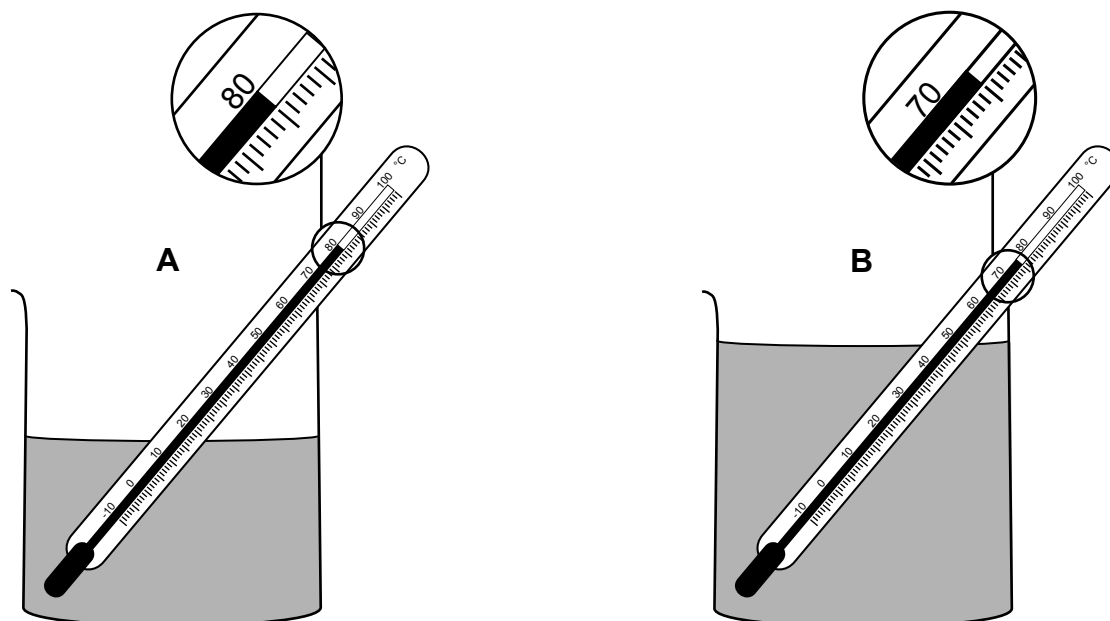


Fig. 3.2

- Approximately 150 cm^3 of hot water was poured into beaker **A** and after a short while the thermometer reading increased to the value shown in Fig. 3.2 beaker **A**.

Read, and record this temperature (at time = 0) in Table 3.1.

- Approximately 200 cm^3 of hot water was poured into beaker **B** and after a short while the thermometer reading increased to the value shown in Fig. 3.2 beaker **B**.

Read, and record this temperature (at time = 0) in Table 3.1.

[1]

- (c) The temperatures of the thermometer in each experiment at times 35 s, 70 s, 105 s, 140 s, 175 s and 210 s are shown in Table 3.1.

- (i) Complete the column headings of Table 3.1 at the dotted lines.

[2]

- (ii) Record the values of time in Table 3.1.

[1]

Table 3.1

time/	temperature of water in beaker A/	temperature of water in beaker B/
0		
.....	76.0	73.0
.....	73.0	71.0
.....	70.5	69.5
.....	68.0	68.0
.....	65.5	66.5
.....	64.0	65.5

- (iii) The results in Table 3.1 show that the temperature of the water in each beaker decreases.

Describe **one** other similarity in the patterns of temperature change of the two volumes of water.

.....
 [1]

- (iv) Describe **one** difference in the patterns of temperature change of the two volumes of water.

.....
 [1]

- (d) Another learner wants to repeat the experiment in order to compare the results.

Suggest **two** factors that should be kept the same in order for the comparison to be fair.

1
 2

[2]

[10]

- 4 Fig. 4.1 shows a leaf.

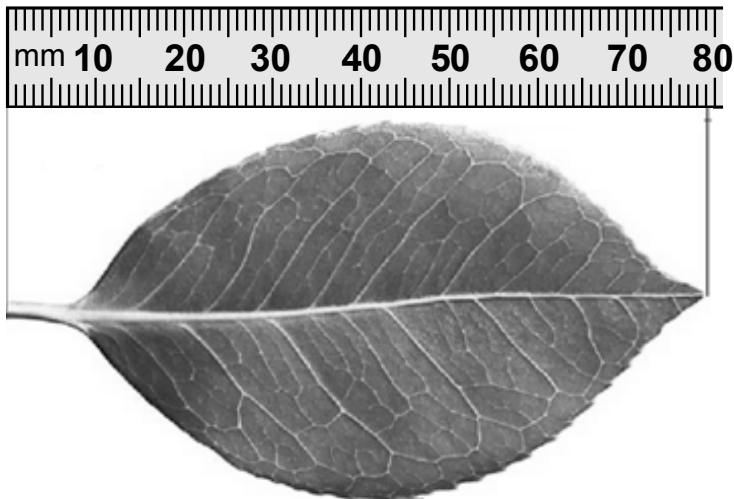


Fig. 4.1

- (a) (i) Make a large drawing of the leaf shown in Fig. 4.1.

[2]

- (ii) Use the length indicated on Fig. 4.1 to calculate the magnification of your drawing. Show your working.

Magnification [2]

BLANK PAGE

Question 4(b) is printed on page 12.

- (b) The lengths of 40 leaves of a plant are measured correct to the nearest millimeter, and the data obtained is recorded in Table 4.1.

Table 4.1

118	127	154	140	136	163	145	135	147	143
146	138	146	161	126	150	180	160	151	131
139	152	120	169	148	162	151	173	130	153
124	170	155	149	149	137	148	144	142	129

- (i) Complete Table 4.2 for this set of data values.

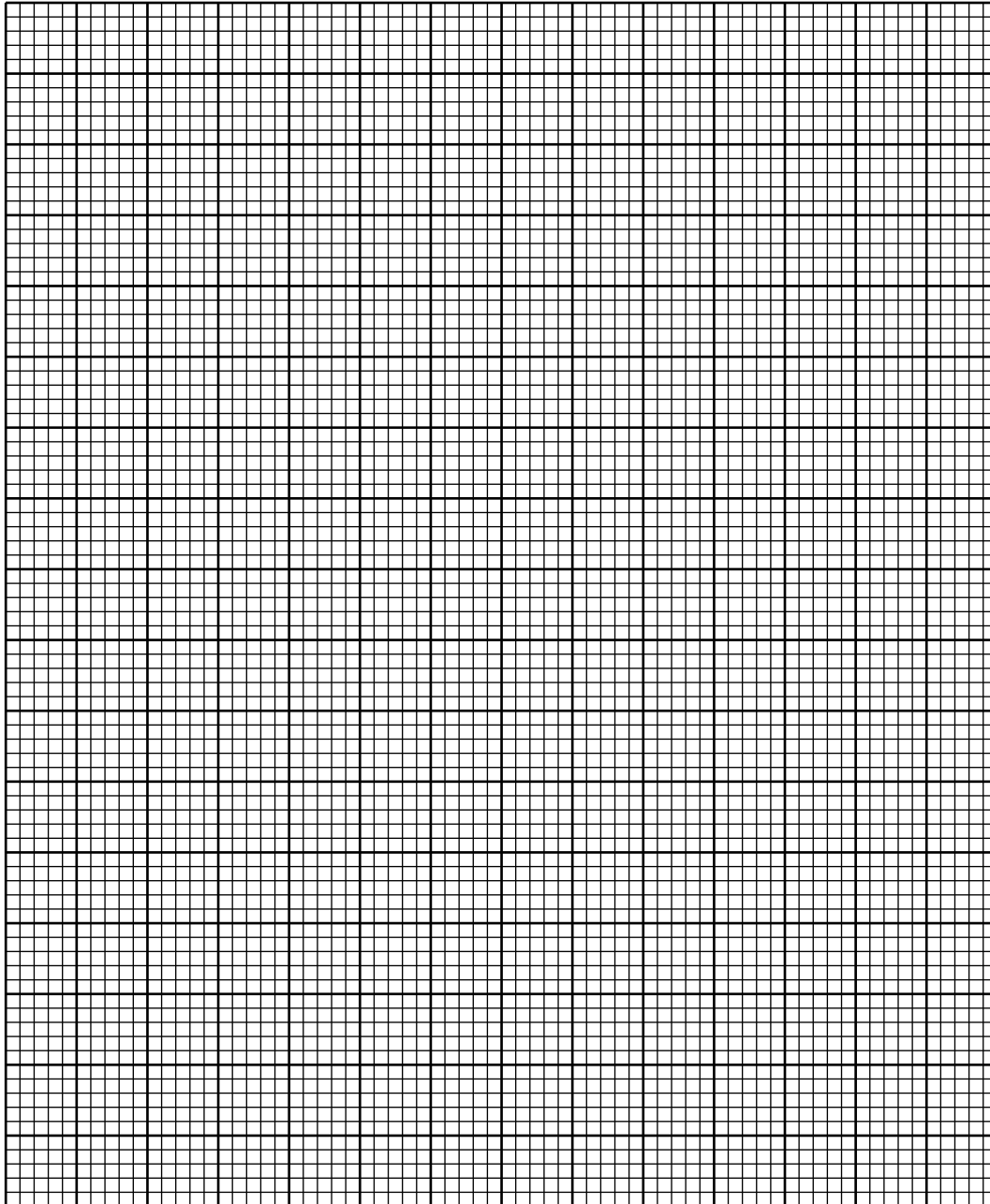
Table 4.2

range	tally	number of leaves
110 – 124		
125 – 139		
140 – 154		
155 – 169		
170 – 184		

[2]

(ii) Draw a histogram of the results shown in Table 4.2 on the grid provided.

For
Examiner's
Use



[4]

[10]

BLANK PAGE

BLANK PAGE

BLANK PAGE

The DNEA acknowledges the usage and reproduction of third party copyright material in the NSSC Assessment, **with and without permission** from the copyright holder. The Namibian Government Copyright Act allows copyright material to be used limitedly and fairly for educational and non-commercial purposes.

The Directorate of National Examinations and Assessment operates under the auspices of the Ministry of Education, Arts and Culture in Namibia.

© MoEAC/DNEA

6116/3/24

345365